Semantic Governance Guide v2.3 — Amendment Set (Draft)

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Document control and how to read this set

This set proposes version 2.3 of the Semantic Governance Standards Guide as twelve amendments to the v2.2 Draft of November 1, 2026. It does not restate the 200-page guide; each amendment names the v2.2 section it changes and gives replacement or added text.

v2.2 already covers much of the thread's design: opaque tenant IDs, lexicons for enterprise and tenant labels, primitives and polymorphic terms, and decision evaluation by the AI circuit breaker. The amendments below close only the remaining gaps, and where v2.2 conflicts with the settled design, that conflict is named.

ID	v2.2 section	Change	Type
A1	3.7 Primitives	Every primitive declares a BFO parent; core terms are immutable	Add
A2	3.7, 3.1	Primitives are minted only at core or a top-level domain, never in a sub-domain	Replace
A3	3.7 Polymorphic terms	Limit and report multiple parents	Add
A4	3.1 / 3.2	Three edge types and a default-deny allowed-edge matrix	Add
A5	3.2 customer tier	Tenant concepts derive from a core or domain class; tenants mint no primitives	Add
A6	3.4 quality gates	Three orphan checks, run per graph	Add
A7	3.8 Labels and Personas	Personas supply alternate labels only; scope placement rule resolves GN-D2	Replace
A8	3.8, 2.7	Pending state and HITL disposition for unresolved terms	Add
A9	3.2 Graph Naming	Per-tenant graphs with no cross-tenant path	Add

ID	v2.2 section	Change	Type
A10	RT-04	MTBH defined by the AI Circuit Breaker v3.0 metrics registry	Replace
A11	3.6	Structural measures for the semantic design	Add
A12	3.6	Dashboards read telemetry; reconcile with the Operations Dashboard	Replace

A1–A6: Primitives, lineage, edges and orphan checks

A1 (Section 3.7, Primitives and Polymorphic Terms): add. v2.2 requires every primitive to be unique and to have one owner, but only core classes and relations are checked for BFO lineage. Add this rule after "Unique": *Every primitive declares the BFO class it specializes, and the build rejects a primitive without one.* Add a second rule: *Published core terms are immutable. A term is changed only by deprecating it with a supersededBy reference and adding its replacement in a core release.*

A2 (Sections 3.7 Owners and promotion, 3.1 Containers): replace. v2.2 says a general primitive the core lacks is added to the domain that first needs it, or to its area container if several do. That allows minting in nested domains and containers. Replace with: *A primitive is minted only in the core or in a top-level domain, only with BFO lineage (A1), and only when no existing primitive covers the meaning. A sub-domain never mints a primitive; it requests one from its top-level parent. A primitive needed by several domains is proposed for promotion to the core, which changes only through a deliberate core release.* The status of area-container minting is left open (see Open items).

A3 (Section 3.7, Polymorphic terms): add. A taxonomy concept or class may derive from more than one parent. Add: *Multiple parents are allowed and are kept to the minimum that the meaning requires. A concept with more than two parents is reported to the data steward. Evolution is additive: a parent may be added, and a concept is never moved to a different parent.*

A4 (Sections 3.1 and 3.2): add. v2.2 lists the relations used inside a domain (skos:broader, skos:broadMatch, skos:exactMatch, core:formalizedBy) but names no edge classes. Add a short section *Edge types*:

Edge type	Meaning	Examples in v2.2	Direction
Derivation	A concept or class is built from a parent	dcterms:subject, core:hasPrimaryTerm, skos:broader, rdfs:subClassOf	Upward to the core
Correspondence	Two concepts name the same or a related thing	skos:exactMatch, skos:broadMatch	Lateral only; never replaces derivation
Lexicalization	A label names an existing concept	skos:altLabel, skosxl:altLabel	Label to concept, one way

Then add: *An allowed-edge matrix lists, for each pair of graph tiers, which edge types may cross. Any edge not listed is rejected at build time (default deny).* This keeps the structure from becoming a graph in which every concept can relate to every other.

A5 (Section 3.2, customer tier): add. v2.2 states that customer assets hold one tenant's labels and local concepts and never an ontology. Add: *Every local concept in a customer asset has a derivation edge to a core or domain class. Its correspondence (match) relation is the lateral link to a concept that names the same thing and never replaces the parent. A tenant mints no primitives; a concept that needs one goes to triage as a core change request (A8). Derivation and match edges are re-checked on every core release, because a core change can make either stale.*

A6 (Section 3.4, quality gates): add. The existing checks cover lineage to BFO and orphan terms for the build. Add three named checks, each run per graph and reported with a count:

1. **Concept derivation:** every concept and class has a derivation path to the BFO root.
2. **Schema binding coverage:** every bound source schema element resolves to a concept.
3. **Label resolution:** every label resolves to exactly one concept in its scope, or is in the pending state (A8).

The target for each is zero failures in a released graph.

A7–A9: Personas, tenants and human disposition

A7 (Section 3.8, Labels and Personas): replace the last sentence of the v2.2 labels rule and add a scope rule. A persona vocabulary supplies alternate labels for existing concepts and nothing else. It defines no concept, relation or class. This resolves open decision GN-D2: a persona label stays on the core or domain term only when it holds for every tenant and has been vetted; otherwise it lives in a lexicon at the narrowest scope that fits.

Label holds for	Where it lives	Graph
Every tenant and persona, vetted	skosxl:altLabel on the term	Core or domain vocabulary
One persona across the enterprise	Lexicon, enterprise tier	Enterprise lexicon
One tenant	Lexicon, customer tier	That tenant's graph
One persona in one tenant	Lexicon, customer tier, linked by core:intendedForPersona	That tenant's graph

A lexicon refers to the concept it labels by identifier only. The reference is one way: core and domain graphs never refer to a lexicon.

A8 (Sections 3.8 and 2.7): add a section, Pending terms and disposition. When a label or concept proposal cannot be resolved to an existing concept in its scope, it does not create a concept. It enters the *pending* state and is routed to the human trustee, whose decision passes the same gate as every other decision (v2.2 Section 2.7). The steward chooses exactly one disposition:

1. **Alternate label:** it is another name for an existing concept; record it in the right lexicon (A7).
2. **Domain concept:** it is new for the domain; mint a concept with a derivation edge (A2, A5).
3. **Core change request:** it shows a gap in the core; open a core change under the core release process (A1).
4. **Reject:** record the reason in the change log.

A pending term may exist in DEV and STG only. A release containing a pending term fails the label resolution check (A6). The age of the oldest pending term is a measure (A11).

A9 (Section 3.2, Graph Naming): add rule 14. *Each tenant's labels and local concepts live in graphs named with that tenant's opaque ID. No edge, import or query path joins two tenant graphs. Tenant graphs may refer to core, domain and enterprise assets, and never the reverse.* This is what keeps persona isolation, fidelity and performance as the number of tenants grows: core and domain ontologies are not loaded with tenant material.

A10–A12: MTBH, structural measures and monitoring

A10 (RT-04, Mean Time Between Hallucinations): replace. v2.2 defines the measure as answers served, or hours in service, between confirmed hallucinations. Replace the definition and add the authority: *MTBH = T_ops / N_h, where T_ops is total operational time and N_h is the number of verified hallucinations or breaker trips in that time. The definition, units and event rules are those of the metrics registry in AI Circuit Breaker: Updated Architecture Design (v3.0); this guide cites that registry and does not restate it.* Keep "answers per hallucination" as a separate, named rate if it is still wanted. The older "mean time between harmful outputs" and "Mean Time to Hallucination" wordings are retired. Update the same definition in Appendix D (circuit breaker configuration, Week 10) and in the Appendix F entry for MTBH.

A11 (Section 3.6, measures): add eight structural measures. The 32 measures (DT-01 to DT-14, RT-01 to RT-18) cover answers and services. Add measures for the semantic design itself, each with a target set at the Gate 1 baseline:

Measure	What it shows	Source check
Correspondence density	Lateral links per concept; flags drift toward a fully meshed graph	A4 matrix
Stem-reuse ratio	Share of new terms built from existing primitives	A1, A2
Impact radius	Terms affected by a proposed core change	Change-impact query
Polyhierarchy rate	Concepts with more than one parent; those over two listed	A3
Domain-minted primitive share	Primitives minted outside the core	A2

Measure	What it shows	Source check
Promotion-backlog age	Age of the oldest pending term and promotion candidate	A8
HITL disposition mix	Share of label, domain concept, core change and reject	A8
Orphan rate per graph	Failures of the three orphan checks (target 0)	A6

A12 (Section 3.6, monitoring): replace. v2.2 describes an Operations Dashboard that reads the change log and model files, and a measured agent assurance proposal. State one pattern: *Dashboards read telemetry events only. Every check that computes a measure emits it as a telemetry event; the Operations Dashboard reads those events and the change log and does not recompute measures from model files.* The structural measures in A11 therefore need an emitting check each, and measures with no emitter are listed as not yet measured.

Open items and Appendix F additions

Five items still need a decision before these amendments are final.

Item	Question	Affects
Area-container minting	Does an operational-area container count as top level for minting primitives, or must they sit in a named top-level domain?	A2, Section 3.1
Persona search ranking	The guide's wording on persona ranking in search must match graph-scoped resolution (A7).	Sections 2.7, 3.8
Simulation sequencing	Where simulation sits against the roadmap and the Candidate entry criteria.	Section 3.3, Appendix C
Council approver	One section lists the approver as open while the roles table names one.	Section 3.4

Item	Question	Affects
Other graph naming decisions	GN-D1 and GN-D3 to GN-D7 are untouched by this set.	Section 3.2

Appendix F gains these entries: **Derivation edge**, **Correspondence edge**, **Lexicalization edge**, **Allowed-edge matrix**, **Pending term**, **Disposition**, **Stem cell** (a primitive in the core or a top-level domain), **Orphan check**, and **Structural measure**. The MTBH entry is rewritten per A10.

To apply these, edit the v2.2 source sections named in each amendment, add the document control row for v2.3, and rerun the build checks, including the new orphan checks.